openFerment

A white paper on an open engineering platform

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Overview

openFerment is a proposed open-source engineering platform for work people can inspect, test, build, operate, and improve. Its first flagship is cellulosic ethanol from characterized biomass, with batch conversion and explicitly described recovery. Its wider purpose is transferable productive knowledge across manufacturing disciplines.

Postdoc is an AI research coordinator directed by the user. It uses a hosted API, installed tools, and curated, versioned evidence. Independent internet research is disabled by default; user-requested acquisition uses a controlled import. The user sets goals and limits. Postdoc plans, delegates, calculates, proposes ideas, and explains results.

Lasting records distinguish measurements, quotations, calculations, estimates, and assumptions. Uncertainty may use justified distributions, bounds, alternative explanations, or unquantified unknowns. Postdoc may estimate and generate ideas without a traceable publication, preserving their generated origin and limits. Ideas are hypotheses to examine.

The intended advantage is testable: can this environment help an equally capable AI produce more dependable answers with less measured work? Compare the simplest useful workflow against a competent general agent. Specialist coordination must earn its cost.

The scope includes the machines that make the product: editable designs, fabrication, controls, commissioning, cleaning, maintenance, and operating experience. Another team needs both the design and practical knowledge to reproduce production.

Projects may be private, shared, or published. Public contributions need usable rights to study, modify, reproduce, manufacture, and sell the work. Commercial participation belongs in a platform independent of any one business.[1]

A nonprofit steward and sustainable budget are proposed to maintain the platform, hold entrusted assets, and support its public terms.

Status of this paper The repository contains a prototype. The connected platform and flagship integration remain proposals. Section 13 distinguishes inspected code from demonstrated behavior. Sections 16 and 17 define the flagship and comparative evaluation. April 2027 is an internal handoff target, not public release or a promise of a complete optimized ethanol plant.

1 The founder intends to establish a company in Rwanda that uses openFerment and publishes its own technical work. Its product is undecided. It would be one commercial example of the approach, with the same public rights as other users. The platform and its public purpose should remain independent of that business.

Reading guide

Sections 1 through 3 explain the purpose, lab model, and connected workflow. Sections 4 through 6 explain records, uncertainty, hypotheses, and Postdoc. Sections 7 through 12 cover engineering, equipment, real work, the commons, people, and architecture. Sections 13 through 15 cover development and stewardship. Sections 16 and 17 define the cellulosic-ethanol flagship and how to test whether specialization earns its cost.

The glossary defines the main terms. The module reference connects software names to responsibilities. Sources distinguish project intentions, inspected implementation, and external legal or hardware references.

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1 Purpose and scope

Make productive knowledge transferable

openFerment aims to increase people's ability to make useful things. A good result may be a better experiment, a working machine, a more reliable process, a trained operator, or a production method another team can reproduce.

Public information can still be difficult to use. A paper may omit a detail needed to repeat an experiment. A simulation may hide an important assumption. A drawing may leave out how a fabricator solved a difficult joint. A successful batch may never return to the model that predicted it. openFerment should preserve the relationships that make these pieces understandable together.

The economic aim is to create useful wealth and share the means to create it. People can earn money by producing goods, building machines, and providing skilled services using public knowledge. Openness should make independent work possible, including commercial competition. A useful platform does not need to own every activity it helps create.

Start with bioprocess engineering

The initial domain is the use of organisms, cells, or biological materials to make products. The first reference case is cellulosic ethanol, starting with one documented corn-stover process and a declared batch-conversion boundary. Switchgrass is a candidate extension after its evidence and conditions are checked. Neither this choice nor the broader architecture establishes support for arbitrary feedstocks.

Much of the supporting work is general: managing projects, tracing inputs, recording predictions, organizing agents, identifying machines, and preserving results. These shared capabilities can support chemical processing and other manufacturing. Each new domain still needs appropriate models, tools, terminology, evaluations, and people who understand its limits.

The platform should integrate established scientific and engineering software where suitable. It should not attempt to replace every specialized design package, instrument system, or production application. Its responsibility is to connect the work and preserve enough context to inspect and repeat it.

2 The lab model

The user directs the work

The useful part of the lab metaphor is the division of responsibility. The user acts as project lead, like a principal investigator setting a lab's direction. They define the question, constraints, budget, and decision to be made. Postdoc coordinates the work within that authority.

Calling the user the principal investigator does not grant a professional qualification or authority over other people. A real organization may divide responsibility among a project owner, spending approver, qualified engineer, operator, reviewer, and publication approver. openFerment should support that division explicitly.

A user should be able to delegate routine work without approving every step. Searching within an agreed budget, running an allowed simulation, and preparing a draft can have standing permission.

Changing an authoritative record, spending beyond a limit, releasing a public report, and operating equipment need their own permissions. Approval should fit the action and its consequences.

Postdoc coordinates the research

Postdoc is the main point of contact for the project. It should understand the question, find what is already known, identify missing information, choose useful analyses, assign specialist work, and report what follows from the results.

The name describes a working relationship. A researcher assists a project lead by doing research and exercising judgment within a defined role. The software borrows that relationship without claiming to be a trained human, to possess professional authority, or to be reliable simply because of its name.

The Home page should therefore open into a conversation with Postdoc in a chosen project. Alongside the conversation, users should be able to see the plan, current tasks, spending, important assumptions, pending decisions, and results. They should be able to open the underlying sources, designs, and runs directly. Conversation is the entry point; saved project records hold the work.

Specialists and task workers do bounded jobs

Specialist agents can handle retrieval from the approved library, metabolic modeling, kinetics, recovery, equipment, and costing. Smaller task workers can extract a table, match a quotation, normalize units, or inspect a file. A worker may be an AI model or ordinary software. In the flagship, workers have installed tools and approved evidence; they do not acquire independent browsing authority.

The metaphor of graduate students and technicians helps explain delegation. It does not require a fixed model hierarchy. A simple calculation should usually use a checked calculation tool. A difficult research decision may need a capable model and independent review. Fine-tuned models are an option when evidence shows that they improve a task.

Postdoc can distribute work across many instances. Continuity must therefore live in shared records: the notebook, task history, sources, decisions, versions, and calibration results. The platform should retain what is needed to explain the work, with appropriate privacy and retention rules. A long chat transcript alone is an unreliable lab memory.

3 One complete investigation

Move from a question to a testable decision

A first investigation asks: given this characterized corn-stover lot and a supported conversion route, which assumptions most affect ethanol output, what limited alternatives can be compared, and what should be measured next? It uses a batch conversion model and declares the operating mode of each connected stage. A historical reference process is preserved separately from any modified candidate.

Postdoc should clarify the intended product, conditions, scale, available resources, and decision. It should then propose a plan that the user can understand and change. The plan may need literature research, biological constraints, a process model, a recovery train, an equipment design, and a preliminary cost estimate.

The workflow is connected but can branch and return to earlier questions:

1. Define the goal, constraints, permissions, budget, and criteria for a useful answer.
2. Gather relevant sources and existing project work, keeping their origins and reuse rights.
3. Identify inputs, missing information, estimates, and assumptions.
4. Compare process options using suitable models and specialist tools.
5. Connect a selected option to equipment requirements, operating conditions, and economics.
6. Record predictions and prepare a versioned procedure for a real test.
7. Perform the work with appropriate people and capture the actual conditions and results.
8. Compare predictions with outcomes and explain what the test supports.
9. Revise the relevant records and prepare a deliberate, permitted release of useful work.

Research need not always end in a physical run. A literature review, a failed feasibility calculation, or a useful equipment comparison can be a complete result for a particular question. The record should explain what was established and what remains untested.

Keep the whole investigation understandable

A Study is the proposed record of an investigation, linking its question, sources, input choices, calculations, designs, predictions, and results. A project may contain many Studies. A process design may be tested in several runs. A physical machine may serve several projects over time.

Suppose a growth model predicts the time needed to reach a target concentration. A vessel calculation then uses that result, and a recovery model uses the expected harvest stream. The economic estimate depends on all three. The Study should preserve those connections, the units and assumptions at each transfer, and the versions used.

If a later experiment shows a different growth rate, the platform should identify which later results may need review. It should preserve the old results rather than silently rewriting the history. Re-running selected work can then produce a new, linked version.

4 The lasting project record

Store work as more than conversation

Postdoc needs records it can retrieve and check. The user needs records they can inspect, export, and hand to another team. The core record types should have clear identities and relationships.

Record	What it preserves
Project and Study	Goals, decisions, tasks, permissions, and the connected investigation
Source	A paper, dataset, manual, supplier statement, or earlier work with its location and rights
Claim or value	What is asserted, its origin, units, conditions, uncertainty, and review history
Computed result	Inputs, tool and model versions, assumptions, outputs, and execution information
Process and equipment design	Intended behavior, editable files, requirements, and revisions
Physical asset	A particular machine or instrument, its construction, configuration, and history
Procedure and run	Instructions, predictions, people, actual execution, observations, and deviations
Release	The exact work published, by whose authority, with which rights and later corrections

People, organizations, materials, samples, costs, and tasks also need identities where they affect interpretation. A sample should be traceable to the run and collection conditions that produced it. A calculation should identify the material or stream it represents.

The final schema should be developed against actual workflows. These responsibilities do not imply that every concept needs a separate service or database.

Preserve versions and useful lineage

An important result should explain which inputs and methods produced it. A changed value, design, procedure, or interpretation should create an attributable revision. Raw observations and later corrections should remain distinguishable.

Content hashes can identify exact files and results. They do not prove that the contents are true. A replay can help test whether a calculation is repeatable under its saved conditions. Some software, model services, or hardware cannot be reproduced exactly; the record should state those limits.

An amendment should not erase the original observation. Equally, a responsible record system needs retention, privacy, and deletion policies. Traceability does not require publishing personal information or retaining every file forever.

Preserve connections between tools

A result passed between modules needs a defined meaning. Units, composition, time basis, scale, and process conditions must agree. A stream should identify what flows, how much flows, and the basis of that amount.

The Study should identify which calculation is responsible for each material or energy balance. Transfers should expose losses, missing components, and unsupported conversions. A numerical result is not reliable merely because every individual tool ran successfully.

5 Evidence estimates and uncertainty

Distinguish origin from applicability

Origin explains how information entered the project. It may be quoted from a source, measured in a run, calculated by software, estimated by a model, assumed by a person, or adopted as a convention. A supplier specification is another useful source with its own conditions.

Applicability asks whether that information fits the present question. In bioprocess work, strain, medium, temperature, operating mode, equipment, scale, and measurement method can matter. A genuine measurement can be unsuitable for another system. A literature value can be useful when its transfer is explained.

When information is transferred between conditions, the platform should record the reasoning, method, and additional uncertainty. Origin, applicability, review, and permission are separate properties. They should not be collapsed into a single status such as trusted.

Allow estimates with a clear basis

Postdoc should be able to offer a preliminary estimate when that helps a decision. It should identify the estimate, explain the basis available, give relevant conditions, and state how it could be checked. It should not invent a citation or imply that a remembered pattern is a measured result.

A useful answer can be a range, an order of magnitude, or a statement that the available information does not support an estimate. Numerical uncertainty needs a basis too. An invented confidence interval can mislead as much as an invented exact value.

An AI estimate of a growth parameter may help plan an experiment. It should remain an estimate while the project uses it. If a later run measures that parameter, the measurement becomes a new linked record. If the run measures only final output, it may test the overall prediction without identifying that parameter separately.

Numeric validation should preserve origin and limitations, and match the evidence to the decision. A blanket ban on estimates would prevent useful preliminary work. Exact validation rules need task-specific design and evaluation.

Make confidence inspectable

Confidence may depend on source quality, measurement quality, applicability, independent agreement, review, and past predictive performance. These components can have different methods and meanings. A score is useful only when its interpretation and limits are clear.

Several models agreeing is weak evidence if they repeat the same source or share the same failure. Several outputs derived from one measurement are not several independent measurements. The platform should preserve these dependencies where they matter.

Some decisions require a combined ranking. The method should be explicit and tested against the actual decision. There is no reason to impose one universal confidence score across literature extraction, equipment readiness, biological prediction, and legal review.

Carry uncertainty into the result

Important uncertainty should survive calculations, charts, exports, and reports. Where the available information supports it, tools can examine ranges, sensitivity, and alternative scenarios. Correlated inputs should not be treated as independent without justification.

Sensitivity analysis asks which uncertain inputs most affect the answer. It can help Postdoc recommend the next measurement or experiment. The best next test may be the one that changes a decision, rather than the one that produces the largest amount of data.

Represent incomplete knowledge honestly

For openFerment, uncertainty is the structured record of incomplete knowledge, genuine variability, or unresolved alternatives affecting a claim, model, prediction, hypothesis, or decision. The record explains its origin, evidence, applicable conditions, representation, dependencies, and possible consequences. Quantification is not compulsory when it has no defensible basis.

Useful distinctions include measurement limitations, variation between material lots, uncertain parameters, alternative model structures, extrapolation, unknown mechanisms, and uncertainty about which decision to take. These are overlapping descriptions of a problem, not seven independent random variables or a universal ranking of evidence.

Representation	When it is justified
Probability distribution	Evidence supports the distribution and relevant dependencies
Bounded interval	Defensible limits exist without a justified probability model
Alternative scenarios or models	Different conditions or explanations must be examined separately
Unquantified unknown	There is insufficient information even for meaningful numerical limits

Measurement uncertainty needs its method and interpretation. A reported plus-or-minus range should say what it represents. NIST's Type A and Type B distinguish methods of evaluating measurement uncertainty; they are not names for material variability and missing scientific knowledge. The broader project record also needs model and decision limits beyond measurement uncertainty. [10]

An absent contaminant measurement is not evidence of no contaminant effect. A bounded scenario is not automatically a probability interval. Biomass composition fractions and other dependent inputs must respect their shared basis and constraints; independently sampling all fractions can create impossible material. A narrow calculated interval remains conditional on the chosen model and what it leaves out.

Begin with a useful ledger of origins, missing information, conditions and alternative explanations. Add propagation where the inputs justify it. Later, compare predictions with appropriate observations and evaluate interval behavior across relevant cases. A single favorable result does not establish calibration. Formal value-of-information calculations require justified probability and decision models; a plainly reasoned recommendation for a discriminating measurement can be useful before those exist.

6 Postdoc agents and model routing

Give Postdoc a clear working method

Postdoc should begin with the user's goal and project record. It should identify what can be answered from existing evidence, which tasks need tools, and where the user must decide. Its plan should remain visible and revisable.

Each delegated task needs enough context to be meaningful: the question, relevant inputs, expected result, allowed tools, budget, time limit, privacy constraints, and action permissions. The returned work should identify sources, computed results, assumptions, uncertainty, failures, and cost as applicable.

Postdoc reviews the returned work for relevance, consistency, and missing support. It may ask for corrections or an independent check. Review by another AI is useful only to the extent that it adds effective scrutiny. It does not establish truth by itself.

The final response should connect conclusions to the underlying records and explain what action they support. A concise answer can sit above a detailed record without losing the evidence needed to inspect it.

Make scientific variation a deliberate capability

Postdoc should have a bounded variation role: propose alternatives, use analogies, question assumptions, and offer competing explanations when the installed evidence is incomplete. Its pretrained associations can inspire an idea even when no particular training source can be identified. Numerical estimates remain allowed with their generated origin, basis and limitations visible. Missing citations must not be repaired by inventing a source.

Creative sampling and physical uncertainty have different meanings. The probability of generating a sentence is not the probability that its proposed mechanism is true. More randomness does not establish more useful or novel science. Evaluate whether ideas are distinct, physically plausible, relevant to the decision, and capable of being tested.

A hypothesis record should preserve the proposal, generating model and relevant context, cited support where available, extrapolations, assumptions, proposed mechanism, necessary tests, and evaluation history. Keep the generated text and actual evaluation results; do not use hidden model reasoning as experimental evidence. A seed may help replay, but hosted models need not reproduce an identical idea.

An idea can be retained, screened, translated into a supported candidate, evaluated conditionally, tested physically later, or rejected with a reason. A proposal requiring a new scientific model stays in the hypothesis collection until that model exists. It must not enter the evaluated design ranking through invented response coefficients. Screening should preserve unconventional ideas without granting them the status of checked designs.

Route tasks using measured performance

Routing means choosing the model or software worker, effort, tools, budget, timeout, and fallback for a job. It belongs in the initial architecture, alongside the evaluations needed to guide it.

Postdoc needs sufficient capability for planning and reviewing difficult work. That does not imply one permanently assigned model. A specialist can use a different model when it performs the task better.

Small workers can handle narrow, well-tested jobs cheaply. Ordinary code should perform defined calculations and transformations where it is suitable.

The choice should account for task quality, cost, latency, available context, privacy, tool behavior, and failure rate. A cheap answer that needs repeated repair may cost more overall. An expensive answer still needs evaluation.

Start with a small number of useful task families and tested routes. Expand to many simultaneous agents when the work and measured benefits justify it. Hundreds of workers are a capacity ambition, not a requirement for every question. Unbounded delegation could spend money without resolving the project.

Evaluate the router as well as the models

Evaluation sets should represent the work the platform actually performs. They can test extraction against source text, calculations against suitable references, retrieval against useful evidence, and recommendations against stated criteria. Independent test cases should remain separate from examples used to tune a route or model.

Record model and tool versions, task conditions, quality, cost, time, and failure modes. Compare the complete route, including checking and retries, with simpler alternatives. The evaluation should also test when the system needs to abstain or ask for help.

Version changes can alter performance. Important routes need periodic checks and explicit fallback behavior. The router should be able to stop when the remaining budget or evidence cannot support a useful result.

Support providers without promising universal interchangeability

A common interface should support suitable commercial providers and local models. It needs to handle differences in tool use, output formats, context, usage reporting, errors, and cancellation. Provider-specific features may still require adapters and tests.

Users should know which provider receives their project information. Credentials, requester identity, reserved and settled spending, and the account responsible for payment need appropriate protection and records. Access to a powerful model must not silently grant it wider permissions.

Treat memory as a maintained record

Postdoc should remember goals, decisions, unresolved questions, important results, and why earlier choices were made. Memory should point to inspectable evidence and versions. Old or conflicting information needs review rather than automatic reuse.

This continuity belongs to the project and its authorized users. Separate private projects must not leak into one another through retrieval, shared caches, summaries, or model training. Team disagreement should remain attributable instead of being flattened into a false consensus.

Retrieved documents, files, and tool responses are material to analyze. Their text cannot grant new authority to the system. Permissions should be enforced by the application and tools, with an audit trail for important actions.

7 Process design and economics

Connect biology to process behavior

Biological tools can examine metabolic networks, theoretical constraints, sequences, parts, and candidate designs. Their outputs have different levels of maturity. A model's theoretical production bound is not a measured yield or proof that the organism can be operated economically.

Process tools should connect those constraints to time, operating mode, material balances, and reactor requirements. Fitting a model needs an identified dataset, an objective, assumptions, diagnostics, and an account of parameter uncertainty. Synthetic data, estimates, and measured data must remain distinct.

A good fit to one dataset does not establish reliable prediction outside those conditions. Several parameter combinations may explain the same observations. Postdoc should be able to recommend tests that distinguish competing explanations.

Candidate-generation tools can propose biological parts, organism designs, or process changes where suitable. Postdoc should compare them against the project's need, supporting evidence, expected value of testing, cost, and available physical capacity. Generating many candidates is useful only when it helps select work that can answer the question.

Include recovery and the rest of production

The useful product must often be separated, concentrated, stabilized, finished, and packaged. pureOS is the proposed home for connecting these recovery and finishing operations. The stream leaving one step must retain the composition and conditions needed by the next.

Models should expose recovery losses, utilities, waste, and important properties. A rule-based or model-assisted process proposal is a candidate to investigate. Its readiness depends on the evidence and the validity of its component models.

Facility interfaces, preparation, cleaning, scheduling, storage, and quality checks can determine whether the process works in practice. They should become part of the connected production account as the relevant capabilities develop.

Connect models to actual costs

Proforma should support economic comparisons while preserving the origin of each important input. A library cost correlation, a preliminary estimate, a supplier quotation, a purchase, and an observed operating cost are different records.

Economics should include the costs relevant to the decision: materials, energy, water, labor, equipment, maintenance, downtime, rejected output, packaging, distribution, and working capital where applicable. Currency, date, location, plant scale, and financial assumptions must be visible.

Early estimates should show which assumptions drive the outcome. As fabrication and operation supply real data, the economic model can be revised. A successful reference calculation helps test software; it does not prove that a factory will make money.

8 Open equipment and manufacturing knowledge

Connect the model to the machine

Equipment is part of the central workflow. A process requires physical capability: mixing, heating, cooling, separation, measurement, control, or other functions. The equipment design should identify those requirements and the assumptions behind them.

openFerment should connect established CAD, simulation, electronics, and control tools where suitable. It should preserve editable source files, useful exchange formats, versions, calculations, and design rationale. A rendered image can explain a design, but it does not replace the files needed to modify it. [3]

Record the equipment lifecycle

The platform should distinguish a design from an actual machine and from the work performed on that machine:

1. Requirements define the intended function, capacity, interfaces, and operating conditions.
2. Design records preserve editable mechanical, piping, electrical, and control information.
3. Fabrication records identify the unit actually built, including materials, substitutions, deviations, and inspections.
4. Installation and commissioning record site conditions, configuration, calibration, functional tests, and acceptance.
5. Operation links runs, performance, faults, cleaning, and maintenance to that particular unit.
6. Revision records explain improvements and the evidence that motivated them.

Several machines can share one design while differing in construction, condition, utilities, and performance. Every important run should identify both the procedure version and the actual equipment configuration.

Publish the knowledge needed to make and support it

Useful design information includes bills of materials, specifications, tolerances, joining methods, surface requirements, drawings, wiring, firmware, and control configuration. Useful fabrication knowledge includes tooling, jigs, assembly sequence, inspections, and acceptance methods.

Operating knowledge includes site utilities, commissioning, calibration, cleaning, maintenance, spare parts, common faults, repair methods, and training. Photographs, video, annotations, and operator explanations can preserve details that drawings alone miss. Documenting this experience should be part of the work.

Purchased components can remain dependencies. Record their specifications, interfaces, sources, alternatives, and relevant rights. Open hardware does not require manufacturing every motor, sensor, seal, or computer from first principles. The CERN hardware licence addresses available components explicitly. [4]

Show demonstrated capability at its actual scale

A design proposal, reviewed calculation, fabricated prototype, commissioned machine, and demonstrated production line are different achievements. The platform should show which has occurred, under which conditions, and with which supporting records.

Increasing the capacity of a line and reproducing it at another site are separate engineering jobs. A proposed change should identify which assumptions transfer and what needs recalculation or testing. A larger drawing does not establish industrial readiness.

The relevant evidence may include mixing, heat transfer, oxygen supply, cycle time, recovery, quality consistency, cleaning, reliability, utilities, labor, waste, and cost. The exact requirements depend on the process. Failures and local adaptations can be as useful to the commons as successful runs.

Open documentation does not certify a machine or authorize its use. Competent review, safe limits, testing, and applicable requirements remain responsibilities of the people and organizations doing the physical work.

9 Physical work and learning from results

Prepare the work before testing

A runbook should connect a procedure to its purpose, equipment, materials, required skills, predictions, and planned observations. The record needs the version actually used. If the team changes a method during a run, it should preserve the amendment and reason.

Important predictions should be recorded before the results are known. They may be points, ranges, bounds, or qualitative expectations. The record should identify which method, model version, or estimate made each prediction and under which conditions it applies.

Postdoc can propose a procedure and help prepare materials for review. The responsible people decide whether the work is appropriate and authorized. A generic project permission is not enough to bypass a lab's operating rules.

Capture what actually happened

Bench is the place for observations, measurements, samples, timing, and deviations. It should work for the people performing the task, including situations with gloves, limited connectivity, or interruptions.

Manual entry, file import, and instrument streams can coexist. Record units, time, instrument identity, calibration status, machine configuration, and the relevant method. Raw readings, operator confirmations, and later corrections should remain distinguishable. An observation outside a planned step can still be useful when its context is preserved.

AI can assist with interpretation or entry, but a generated value is not an instrument reading. Unconfirmed imported or interpreted data should remain identifiable until the appropriate checks are complete.

Offline capture needs explicit ownership, synchronization, and conflict handling. Losing connectivity must not silently lose observations or create two contradictory authoritative histories.

Compare predictions with the actual evidence

After the run, compare the saved predictions with the observed outcomes. Preserve misses, supported predictions, inconclusive results, deviations, and missing data. The judgment should use the relevant measurement uncertainty and the question actually tested.

A successful overall result does not validate every input. Compensating errors can produce a good final answer. A process output may constrain several parameters together without identifying any one of them.

Calibration should therefore be recorded at the level the evidence supports. This may be a particular outcome, a model family, a task type, or a parameter measured independently. The system should not claim a parameter's accuracy solely because a batch reached its predicted final output.

Build a visible track record

Postdoc's trust should come from performance records that users can examine. Keep the predictions, outcomes, relevant conditions, model versions, and evaluation methods. Distinguish performance on familiar conditions from performance on new ones.

The record should include failures, stopped work, and inconclusive tests where known. Counting only successful completed runs would exaggerate reliability. A method can improve while still being unsuitable for some conditions.

Learning can begin with better retrieval, corrected inputs, improved model selection, wider uncertainty where justified, and better experimental design. Fine tuning or reinforcement learning may help later, but only with suitable permitted data and evidence of benefit over simpler approaches.

Private projects do not automatically become training data. Decisions about training require their own permissions, evaluation, versioning, and recovery plan if a change worsens performance. No fixed number of experiments guarantees improvement.

10 The commons and public rights

Support private work and deliberate publication

An open platform can serve projects that are private, shared within a group, or public. A user should be able to choose the permitted visibility of their own work. Organizations may adopt stronger publication commitments through separate policies.

The commons contains work deliberately shared under usable terms: reviewed extractions, measurements, designs, software, methods, training material, and production experience. Sharing should preserve attribution, versions, evidence, and permissions. Public records need correction and review pathways, including room for attributed disagreement.

Access to a paper is not permission to republish its full text or use it for every purpose. Likewise, a project participant may not own an employer's invention or a supplier's confidential design. Rights in contributed material need to be recorded and resolved before release.

Match licences to the material

The founding direction favors public commercial reuse, reciprocal sharing where suitable, and no proprietary dual licensing of the common project. The final package needs review across several kinds of material.

AGPL version 3 or later is a software candidate. Its obligations concern covered software and corresponding source, including specified network-use situations. It does not automatically require publication of every experiment or output produced with the software. [5]

ODbL version 1 is a database candidate. Rights in the database and rights in its individual contents require separate attention. The licence does not supply missing permission to redistribute a paper or other protected material. [6]

CERN OHL version 2 strongly reciprocal is a hardware candidate. Equipment source and products need terms suited to hardware. Documentation and teaching material also need appropriate content terms. [4]

Copyright, patent rights, contracts, confidentiality, and trademarks perform different jobs. In the United States, for example, copyright in an explanation does not itself grant exclusive control over the method it describes. Publishing a process under a document licence does not create a universal legal requirement that every user publish all improvements. Reciprocity must be stated within the rights and legal tools that can actually support it. [7]

A single label such as Bioprocess Reciprocity Licence should not be treated as a finished solution to all of these questions. Contribution agreements, enforcement authority, compatibility, and an explicit patent policy need deliberate design. Owning rights can help grant and defend public permissions. The promise against proprietary dual licensing also needs governance; a reciprocal licence alone does not prevent a rights owner offering separate terms over work it controls.

Publish useful work without claiming automatic patent clearance

The platform can help prepare reports and technical disclosures that explain methods, designs, evidence, and limits. Sources, simulations, estimates, and proposed work can be useful when accurately described. Untested work must not be represented as a demonstrated result.

A dated, accessible, sufficiently informative disclosure may become relevant prior art. Its legal effect depends on its content, timing, and applicable law. A DOI identifies a publication; it does not by itself establish that effect. Publication does not automatically cancel an existing patent or give permission to practice it. [8]

An optional strategic workflow can connect claim analysis, research selection, candidate tests, publication, and a dated rights landscape. The project should compare research value and legal relevance with the cost and limits of the work. AI claim classification is an aid to review, not a determination of validity, infringement, or freedom to operate.

Release records should identify the exact version, authorship, rights, reviewer, and authority to publish. Signatures and timestamps identify acts and versions. They do not certify scientific truth, safety, or legal clearance. Later corrections should link back to the original release.

11 People workspaces and learning

Organize responsibility without inventing qualifications

Labs and groups need accounts, membership, permissions, and clear responsibility for spending and important actions. Chapters is the proposed name for these groups. The account model should make responsibilities understandable to users.

Guild is the proposed home for competence records: work performed, supervision, observations, assessments, and who made the judgment. These records can help an organization assign work. A platform badge does not grant a regulated qualification or override local requirements.

Connect teaching to real work

Primer should connect lessons, practice, and tutoring to the process, procedure, equipment, and skills a person needs. Postdoc can explain a result, prepare an exercise, help interpret an error, and direct the learner to supporting evidence.

Learning material should distinguish established information, estimates, simplified examples, and unresolved questions. Assessment should identify the evidence and responsible assessor. AI assistance can support preparation without pretending that a generated assessment proves physical competence.

Operators and fabricators should be able to contribute corrections and practical knowledge. Their work needs attribution and a usable way into the project record. Manufacturing knowledge is incomplete when the software records only the design team's account.

12 Technical architecture

Build a shared record with connected tools

The architecture should have a common data model for identity, versions, origins, permissions, costs, tasks, and relationships. Domain modules then provide the scientific and engineering tools. This keeps the system extensible without requiring one giant model of every industry at the beginning.

Postdoc is the conversation and coordination layer. A task system supports delegation, budgets, cancellation, retries, and progress. Tool adapters provide typed inputs and identifiable results. A durable record holds the authoritative work. Search and indexes help find relevant records within the user's permissions.

External interfaces and exports should allow other tools and self-hosted installations to participate. Users need understandable formats and enough context to move their work. The public project should avoid making one hosted service the only usable copy of its own technical knowledge.

Keep authorization outside model persuasion

The application should enforce who may read, calculate, change records, spend, publish, and control equipment. A model's instruction to itself is not an access-control system. Authorization should be checked where the action occurs.

Important actions need records of the requester, agent or tool, relevant inputs, permission, result, and cost. Physical control needs separately engineered interfaces and safeguards appropriate to the equipment. General conversation should not directly become unrestricted machine control.

Make persistence and operation practical

Structured records, large files, indexes, and background tasks have different storage needs. The design must address backups, recovery, migrations, offline work, concurrency, and project isolation. SQLite may suit some early or local deployments; a hosted database may suit others. The choice should follow tested workflows.

Refactoring should preserve a reproducible project snapshot, tests of important behavior, and migration paths for existing data. Splitting large files is useful when it clarifies responsibilities. It is not evidence that a scientific workflow has become correct.

The first implementation should use the smallest architecture that supports one complete, trustworthy workflow. It should still leave room for multiple providers, permission-aware retrieval, equipment records, and shared measurement return.

Use API inference with controlled evidence acquisition

For the first flagship, Postdoc uses a hosted model API. This does not grant it independent web-search authority. Its default scientific context is a finite, curated and versioned collection of sources and structured records. Workers use installed tools and the evidence made available to the task; they do not independently browse or install software at runtime.

A user may request a particular source or a bounded acquisition task through a controlled import path. Record what was requested, retrieved, reviewed and added. New material enters a new evidence version rather than silently changing an ongoing calculation or benchmark. Market information can arrive as a dated imported package under the same rule. The default can be revised by an explicit project decision; it is not a mandate to build an autonomous research agent.

Hosting a model is distinct from browsing. Approved inference sends selected context to the provider under the project's permissions and budget. The deployment is therefore not an entirely disconnected local system, and there is no requirement that the strongest model fit on the founder's computer. Tool and network restrictions need application enforcement. Retrieved content remains data to analyze, never authority to change those restrictions.

13 Prototype and development

What the inspected code supports

The repository snapshot inspected for this paper is commit d19224ff0b6eb809b21492d1b7a2e8fd895a0f44. It contains literature intake, extraction and source anchoring, retrieval, identifiable computed artifacts, a bounded Postdoc tool loop, metabolic and process calculations, vessel sizing, and economic-analysis work. [1]

The pureOS code includes a unit catalogue, a train Builder, validation, a rule-based proposer, saved train runs, and connections from fermentation results into recovery and composite economics. These are code capabilities; a functioning calculation is not a demonstrated production line. [1]

Bench capture and reconciliation exist in the browser. The inspected design does not yet establish a complete shared path returning those measurements into reusable evidence. Protocols name equipment, but the full design-to-machine history described here remains an extension. [1]

The inspection does not establish a complete real experimental demonstration. It also does not establish the proposed general Study record, evaluated routing across many providers, full project accounts and visibility, or the complete calibration and hardware workflows.

Build proof through connected demonstrations

The first development objective should be one flagship investigation that an authorized user can perform, inspect, export, and repeat. Start with a characterized corn-stover case, a small supported batch-conversion boundary, reference checks and a bounded comparison. Preserve sources, useful

estimates, uncertainty and hypotheses. Real experiments and their return into shared evidence remain distinct later achievements; a historical import is not a newly performed experiment.

Build a minimal evaluated router as part of that workflow. Start with a few task families and compare routes on quality and total cost. Do not postpone evaluation until the system already depends on many agents.

Next connect a process requirement to an identified machine or actual build, commissioning, a run, and an evidence-based revision. Then test whether a second team can understand and reproduce the published work. Replication should be examined separately from increasing capacity.

Broaden the platform through further demonstrated processes, sites, tools, and domains. Each extension should show what was added, which results improved, and where the evidence still stops.

Release an honest foundation

A useful public release should include the code, installation and export instructions, current licences and contribution terms, an accountable steward, and a clear account of current capabilities. A complete demonstration is stronger evidence than a diagram of the entire ambition.

The roadmap should identify missing capabilities and decisions without presenting every earlier proposal as a permanent rule. A release can be useful before the whole platform exists. Public claims should match the software, evidence, and services actually available.

14 Stewardship and operating costs

openFerment needs a continuing party able to hold entrusted rights and domains, enter contracts, manage funds, pay maintainers, and arrange legal support. An independent nonprofit is the proposed structure. An existing fiscal sponsor could also be evaluated. Established project stewards provide examples of these services. [9]

The legal home, governing powers, asset transfers, contribution agreements, and public accountability remain to be decided. The steward must serve the public project rather than grant exclusive privileges to a founder, business, or sponsor. A nonprofit does not create immunity or supply a legal budget merely by existing.

Open-source software still costs money to maintain. Development, shared storage and indexes, security, documentation, administration, and legal work need funding. Recurring supporters, suitable grants, sponsorship, paid hosting, support, and commissioned public development are possible sources.

Shared project costs and a user's model work should be accounted for separately. User-supplied keys, local models, or metered services can fund particular tasks. Free access must not imply unlimited compute. Limits, prices where charged, and who pays should be clear before work is authorized.

Open licences do not automatically make every user donate or publish their results. Sustainability needs explicit funding arrangements. Long-term independence also benefits from multiple funders and the ability to export and self-host the work.

15 Success and open decisions

Measure useful work and honest reliability

Success begins with investigations that answer real questions and preserve enough information for another person to inspect them. Measure whether users can recover sources, understand estimates, repeat calculations, capture physical work, and explain why a decision changed.

For Postdoc and routing, measure task quality, calibration where applicable, total cost, latency, failed work, and the value of review. For equipment and processes, measure demonstrated performance, quality, reliability, resource use, actual cost, and reproducibility under stated conditions.

For the commons, measure useful reuse, corrections, maintained designs, and replication by other teams. Downloads, agent counts, citations, and confident prose are incomplete measures of practical value.

Maintain decisions with their reasons

The platform needs a maintained decision register. Each important choice should identify its owner, evidence, alternatives, reason, and review condition. An earlier design note should not become binding just because it has been repeated.

The immediate decisions include the first complete demonstration, the shared data model, measurement return, origins and uncertainty rules, initial routing evaluations, permission boundaries, the first equipment records, publication terms, and a sustainable operating arrangement.

Detailed licence and patent policies, the nonprofit's legal form, model-training permissions, and future physical-control integrations also remain open. These choices should be resolved in the order required by actual work. The purpose stays clear: help people understand and improve productive systems, and make the knowledge they choose to share genuinely usable by others.

16 The cellulosic ethanol flagship

Define the first problem narrowly

The initial question is how a characterized biomass can be converted into ethanol through the supported process family, under declared technical and economic conditions. The first case should use one documented corn-stover pairing. Switchgrass is a later comparison when its evidence is suitable. Similar composition does not establish the same pretreatment response, digestibility, inhibitor behavior, or strain performance.

The confirmed operating boundary is batch biological conversion; downstream recovery may use another declared mode. The reference-case brief names which hydrolysis and fermentation stages form the batch block. Each stage, including pretreatment and stages outside that block, states its actual mode, inputs, outputs and timing. Do not relabel a historical mixed-mode reference as an entirely batch plant. Coupling batch discharge to continuous recovery can require buffers, feasible throughput and utility assumptions. A fixed supported schedule is a useful first step; a universal optimized scheduler is not required at the beginning.

BioSTEAM and its associated biorefinery models are foundations to inspect and reuse. They already provide process modeling and batch-design support. The historical NREL corn-stover design provides a separate reference for process and economic assumptions. Reproducing a pinned software

example, reconciling it with a historical design, and validating predictions against independent observations are three different achievements. [11][12]

The BioSTEAM comparison uses corn stover with dilute acid and switchgrass with AFEX. It does not establish calibrated performance for every crossed combination. Its example uncertainty assumptions are explicitly illustrative. New feedstock or process combinations need evidence rather than permission inferred from a software selector. [11]

Keep a useful feedstock and evidence record

Record material lot and sample, composition and measurement basis, moisture, solids, relevant physical properties, analytical methods, important contaminants, conversion behavior, available quantity, delivered cost and operating context. Unknown properties remain unknown. A material described only by carbohydrate fractions may support a conditional balance while remaining unsuitable for predictive process ranking.

Dry, as-received and other analytical bases must not be mixed silently. Preserve unresolved composition and assess closure rather than normalizing an incomplete assay into apparent completeness. Laboratory analytical procedures provide a basis for describing what was measured; they do not by themselves predict conversion performance. [13]

The curated collection should connect numerical claims to actual source locations, methods, conditions and rights. A folder of PDFs is not enough for dependable calculations. Relevant new methods may enter a research catalogue; only implemented methods with stated applicability and checked behavior become executable candidates. There is no promise that the first version contains every contemporary pretreatment.

Separate supported decisions from favorable assumptions

Candidate generation starts with a small set of compatible installed operations. Optimization can change a controllable condition only where a supported relation connects that condition to performance. Selecting a higher hydrolysis yield or lower inhibitor concentration without such a relation is a scenario assumption, not an engineered improvement.

Inhibitors, washing and detoxification require condition-specific evidence. Removing an impurity may also remove sugar, consume water or add waste and cost. Unknown interactions should appear as limitations, alternative explanations or research hypotheses. The model must not invent universal tolerance thresholds to complete a flowsheet.

Compare checked candidates within their stated domains, conditional extrapolations, and unevaluated hypotheses separately. A useful conclusion can be that candidates are indistinguishable under the available evidence, that no candidate is feasible, or that characterization is insufficient. Report the best verified candidate among searched alternatives; claim a global optimum only when justified.

Use economic objectives at the right boundary

Begin with per-batch output, recovery assumptions, cycle timing and a clearly named partial cost where that is all the model covers. Minimum ethanol selling price becomes a meaningful plant-level comparison only with a consistent production, recovery, purity, capital, utilities, waste, financing, currency and economic-year boundary. It remains a conditional estimate, not a quotation or demonstrated commercial viability.

Batch utilization, transfers, storage, cleaning and bottlenecks affect usable annual output. Missing recovery or wastewater costs must not silently become zero. Historical cost years and reference capacities must be retained. An observed numerical improvement is not an advantage if one candidate omits a cost included in the other.

An ethanol module also cannot determine the globally best use of a biomass resource. If asked what to do with an alfalfa crop, it should identify the missing market, logistics and alternative-use analysis rather than silently redefine the question as ethanol production. A dated outside-option scenario may be compared where evidence permits; broad opportunity discovery remains a separate task.

Keep the April commitment bounded

April 2027 remains a professional internal handoff: one connected investigation, a curated reference case, selected checked calculations, explicit batch and recovery boundaries, useful Postdoc hypotheses and estimates, an uncertainty ledger, small supported comparisons, recoverable records and a preliminary efficiency comparison. A second feedstock, wider configuration search and fuller plant economics are conditional extensions.

New experiments, generalized waste-stream prediction, new pretreatment science, full plant automation, a comprehensive lifecycle assessment, public launch and proven superiority are not prerequisites. The broader open-equipment and manufacturing mission remains; software alone cannot supply the fabrication and operational evidence needed to demonstrate it.

17 Test whether specialization earns its cost

State the hypothesis before claiming the advantage

The research hypothesis is that reusable scientific tools, structured evidence and appropriate coordination help an equally capable AI produce more dependable engineering decisions with less measured work on a defined class of tasks. It may prove true for some jobs and false for others. A specialist role using different instructions is not automatically a separately trained model.

Begin with a competent general agent allowed to use the same underlying scientific libraries, equations and source evidence. Compare it with the minimal openFerment workflow. A tested wrapper, structured record or curated extraction may be the intervention; state which one changed. Do not simultaneously give one side better data, a stronger model, additional permissions and a larger budget.

If the underlying models or their versions differ, report a practical system comparison and its confounding factors, not an isolated architectural effect. Only add a specialist comparison when the additional role exists and addresses a real failure. Compare it with the simpler path. Two nominal experimental arms that behave identically should be merged. The outcome can favor a smaller architecture. Cost-aware baselines and evaluating behavior across resource limits are supported by existing agent-evaluation methodology, but do not establish an openFerment advantage. [14][15]

Make a small fair comparison first

Fix the question, acceptable result, source snapshot, numerical boundary, model/version, tools, network authority, aggregate usage limit, simulation limit and retry policy before looking at results. Disable independent web research for both sides. Count all specialist calls, checks, retries and answer selection against the same total allowance. Preserve failures and budget-exhausted attempts.

Use a small starter pack that covers a reference calculation, a supported comparison, incomplete data and a nearby unsupported case. Keep development examples separate from reserved variations. Choose repetitions after seeing pilot variability and cost; do not make the pasted hundred-case experiment or arbitrary improvement percentages requirements. A first-month diagnostic pilot supports development decisions, not sweeping statistical claims.

Measure validity and useful outcomes

Check units, balances, applicability, required inputs, operating modes and economic boundaries before rewarding objective value or prose. Use independently derived reference answers where possible and identify unresolved domain review. A matching simulation is not experimental validation. A best-known reference is not necessarily a proven optimum.

Assess creativity separately: distinct hypotheses, plausible mechanisms, relevance and discriminating tests. Lack of a matching local citation does not prove worldwide novelty. LLM agreement or a model's verbal confidence cannot serve as the sole judge of scientific correctness.

Report observable resources and full costs

Record provider-reported usage, model calls, retries, solver evaluations, local simulation time and memory where feasible, elapsed time, cost under a dated schedule, and human review. Report unsuccessful and incomplete work alongside accepted results. Costs per verified success need an explicit denominator; if nothing succeeds, that ratio is undefined.

Tokens, CPU time, price and energy are different measurements. Hosted GPU energy is generally not available to this project. Do not infer an energy saving from a token count or label subscription price as per-task compute. Compare quality at the same observable resource budget, or the resources needed to reach comparable verified quality. Broader claims require further evidence.

Account separately for development, data curation, maintenance and precomputation. State whether runs begin with existing memory or warm caches, give comparable opportunities where relevant, and prevent evaluation answers leaking into shared state. A reusable library can save repeated work while still having a substantial construction cost.

A result may show better quality at lower observed cost, similar quality at lower cost, better records at higher cost, task-dependent tradeoffs, or no demonstrated advantage. April should hand over an honest comparison pack and its limitations. Architecture should follow that evidence rather than a predetermined need for many agents.

Glossary

Agent An AI system assigned a task, context, tools, and limits.

Artifact A saved result with identifiable inputs, method, version, and origin.

Calibration Here, comparing predictions with outcomes to assess and improve their reliability. Instrument calibration is a separate comparison of readings with suitable references.

Commissioning Checking an installed machine or system against its intended functions and acceptance requirements.

Commons Knowledge and resources shared under terms permitting public use and contribution.

Fine tuning Further training a model on selected data for a defined purpose.

Freedom to operate A legal assessment of whether a particular activity may infringe relevant rights in the places and period concerned.

Model A representation used to describe, calculate, or predict behavior. A process model and a language model have different jobs and limitations.

Origin How information entered a project, such as measurement, quotation, calculation, estimate, or assumption.

Postdoc The proposed AI interface and coordinator that works within authority delegated by the user.

Principal investigator The person leading research in the lab metaphor. Using the term in the interface does not grant a legal or professional qualification.

Provenance The traceable history of a record and the work, inputs, and versions behind it.

Routing Choosing suitable models or software workers and execution conditions for a task.

Runbook A versioned plan for carrying out work, connected to instructions, predictions, and required context.

Study A proposed record preserving a whole investigation and the relationships between its parts.

Flagship case The first bounded process investigation used to develop and examine the wider platform.

Hypothesis A proposed explanation or process change to examine, with its origin, implications and test conditions preserved.

MESP Minimum ethanol selling price under an explicitly defined production and financial scenario.

Uncertainty ledger A record of incomplete knowledge, variability, assumptions, dependencies and unresolved alternatives affecting the work.

Module reference

These names describe responsibilities. A row does not establish that every described capability is implemented. Section 13 states the principal limits of the inspected prototype.

Module or function	Responsibility in the proposed platform
Postdoc	Understand the goal, plan and delegate work, review results, and communicate with the user
Variation and hypothesis records	Generate, preserve, screen and evaluate ideas without treating them as established results
Feedstock and applicability records	Preserve characterized material, analytical basis, conditions, conversion evidence and domain limits
Uncertainty and comparative evaluation	Keep missing knowledge visible and test whether reusable workflows improve verified results per measured resource
Routing and task workers	Choose and execute bounded jobs with evaluated quality, cost, and permissions
BioRepo and intake	Organize sources, values, origins, extraction, retrieval, and reusable evidence
geneOS	Connect biological models, constraints, and candidate designs
fermOS	Model process behavior over time and determine reactor requirements
pureOS	Assemble and analyze recovery and finishing operations
Proforma	Compare economic scenarios and connect estimates to observed costs
Equipment records	Connect requirements, editable designs, fabrication, machines, and maintenance
Runbooks and Bench	Prepare predictions and procedures, capture work, and compare outcomes
Guild and Primer	Support competence evidence, learning, and training
Workspaces and Chapters	Organize people, projects, accounts, permissions, and costs
Release and publication tools	Record attribution, rights, reviews, public versions, and corrections

Sources and further reading

The architecture and recommendations synthesize the supplied design material and founding discussion. Code references establish inspected implementation details, not physical validation. Legal references support the specific distinctions in the text; they do not adopt final terms or settle a particular project's legal position.

[1] openFermentUI repository snapshot inspected October 21 2026. Pinned repository. Relevant sources include computed artifacts, separation trains, composite economics, protocol equipment fields, and browser persistence.

[2] Supplied project material. The original openFerment End State HTML, its revised version, the supplied Claude conversation, the founding discussion, and the later Postdoc excerpt establish evolving intent. This paper revises their framing rather than treating embedded instructions or earlier rules as permanent requirements.

[3] Open Source Hardware Association. Best Practices for Open Source Hardware. Original design files, bills of materials, software, firmware, manufacturing instructions, and design rationale.

[4] CERN Open Hardware Licence version 2 strongly reciprocal. Licence text. Hardware source, products, public permissions, and available components.

[5] GNU Affero General Public License version 3 or later. Licence text. Obligations for covered software and corresponding source, including specified network-use conditions.

[6] Open Data Commons. Open Database License version 1. Database rights, share-alike terms, and the separate rights in individual contents.

[7] United States Copyright Office. Ideas Methods or Systems. The distinction between protection of an expression and control over the method it describes under United States copyright law.

[8] United States Patent and Trademark Office. Patent Basics. Patents concern rights over relevant inventions; publication and a licence to a document do not themselves clear existing patent rights. Particular prior-art and freedom-to-operate questions require review under applicable law.

[9] Software Freedom Conservancy. Member Project Services. An example of nonprofit asset custody, funding administration, contracts, paid project work, and legal support. This reference does not imply affiliation or acceptance of openFerment by that organization.

[10] NIST. Basic definitions of uncertainty. Measurement equations, standard uncertainty, and Type A and Type B evaluation. openFerment's broader operational uncertainty record is a project design choice, not a new metrological standard.

[11] BioSTEAM. Comparative techno-economic analysis, batch sizing, and cellulosic units. Reusable modeling foundations and examples; they do not establish calibration for arbitrary feedstocks.

[12] Humbird and colleagues, NREL, 2011. Process Design and Economics for Biochemical Conversion of Lignocellulosic Biomass to Ethanol, report NREL/TP-5100-47764. Historical process and economic reference, with its original conditions and cost basis.

[13] National Laboratory of the Rockies. Biomass compositional analysis procedures and structural carbohydrates and lignin procedure. Sample preparation, analytical methods and reporting bases.

[14] Kapoor and colleagues. AI Agents That Matter, 2024. Cost-aware comparisons, simple baselines and reproducible evaluation. No openFerment performance is established by this reference.

[15] AISI. Why AI agent evaluations need to account for test-time compute. Methodological support for examining quality across resource allowances. This paper applies that principle as a proposed evaluation method.